

POWER CONSUMPTION FORECASTING USING PROPHET AND ARIMA MODELS

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ABSTRACT

This study compares Facebook Prophet and ARIMA for 30-day power consumption forecasting at CvSU Imus Campus, finding Prophet outperforms ARIMA with MAPE of 4.2% vs. 7.8%, especially for seasonal patterns.

INTRODUCTION

Accurate power consumption forecasting enables universities to negotiate better electricity tariffs, plan load shedding, and improve energy efficiency. CvSU Imus pays approximately PHP 2.8M annually in electricity. A 5% prediction accuracy improvement could translate to PHP 140,000 in cost avoidance. This study compares classical (ARIMA) and modern (Prophet) time series models for campus power forecasting.

METHODOLOGY

Daily electricity consumption data (kWh) for CvSU Imus Campus from January 2020 to December 2022 (1,096 data points) was obtained from the campus facilities office. Data preprocessing: interpolation of 17 missing values, outlier capping at 3σ . ARIMA(2,1,2) was selected via AIC minimization. Prophet was configured with yearly and weekly seasonality, Philippine holidays as regressors, and COVID-19 lockdown period as a special event regressor. Walk-forward validation: monthly 30-day forecasts over 2022.

RESULTS

Prophet achieved MAPE of 4.2% vs ARIMA's 7.8% on the 2022 validation period. RMSE: Prophet 182 kWh vs ARIMA 341 kWh. Prophet outperformed most during semester start weeks (+82% accuracy lift), which correspond to high seasonality events. Both models underperformed during typhoon-induced power outages.

DISCUSSION

Prophet's superior handling of multi-seasonality and holiday regressors makes it better suited for academic institution energy forecasting where class calendars strongly drive consumption. ARIMA remains competitive during stable mid-semester periods. Future work should incorporate weather features (temperature, solar irradiance) and test LSTM-based models for further accuracy gains.